

Connector Types

Adapters

- RO 11
- RJ 45
- F type → coax
- Lucas connector (LC)
- Straight tip
- Submini connector (SC)
- Punchdown block
- micro usb
- mini usb
- USB C

molex → IDE

Lightning ports

fiber optic

Peripheral Cables

USB Universal Serial Bus

127 devices can be plugged into USB

(USB Hub)

USB → A, B, C

Type A → host port

Type B → guest port

Type C → host and guest port

Thunder bolt

↳ Fastest

Serial Port RS-232 DB9

Hardware

Motherboard

How the computer work?

RAM Random Access Memory

DDR (Double Data Rate) → new

SDRAM → old

Performance Rating

DDR-266 = PC-2100

266 x 8 = 2128

Transfer Rate bit Mbps

Installing RAM

RAM Types

volatile

ROM non volatile

SRAM → cache inside CPU

static

DRAM dynamic

RAM

Dimm Dual In-line Memory Module

SODIMM small outline Dual In-line memory module

RAM Channeling

single vs Double sided

triple vs Quad

more expensive

RAM type is most important

RAM variations

- # Latency
 - ↳ CAS level / in milliseconds
 - ↳ lower is better for either
- # ECC (Error checking and correcting) RAM
 - ↳ more stable
- # Registered and Buffered Memory

Ram considerations

Hard Drives

- Capacity
- Performance
 - ↳ Throughput
- Interface
 - ↳ Internal (SATA or m.2)
 - ↳ External (USB or Thunderbolt)
 - ↳ Also known as connectors
- Form Factor
 - ↳ dimensions of the disk

HDD Hard Disk Drive

cheaper

Rotation Per Minute (RPM)

for Data storage

not for Operating System

Other Storage Media

- # External Drives
 - ↳ convenient
- # Flash Memory Devices
 - ↳ USB Thumb / Flash / Pen drives
- Memory Cards
- Card Reader
- Optical Drive
 - ↳ CD
 - ↳ DVD
 - ↳ Blu-ray

RAID system

Redundant Array of Independent Disks

Data protection

- Hardware RAID
- Software RAID

JBOD (Just a Bunch of Disks)

RAID 0

↳ Faster Speed

disk striping

No Data protection

RAID 1

Disk Mirroring

Fast recovery

RAID 5

Parity data

three disks minimum required

Parity slow

RAID 10 (1+0)

super fast

both RAID 1 & RAID 0

Four Disk minimum

- * IF RAID 1 fails
 - ↳ all data will be lost
 - ↳ but two disks in a single RAID 1 fail.

SATA → connector

Mini SATA (mSATA)

eSATA

SSD Solid State Drive

available in SATA and M.2 form factors

* Limited writes cycles

m.2 / NVMe SSD

PCIe or SATA

M.2 connectors :

NGFF (Next Generation Form Factor)

→ keyed B key, M key, B+M key

NVMe (Non Volatile Memory Express)

M.2 22110

M.2 2280

millimeter form factor

Cooling Components

Throttling & Overclocking

Intel → Speed Step

AMD → "Cool'n'Quiet"

Clock Multipliers

* CPU clock × Clock multiplier = CPU frequency

* 150 MHz × 10 = 1500 MHz / 1.5 GHz

Liquid Cooling

Passive Cooling

- Thermal Compounds
- Thermal Pads
- Heat Sinks
 - ↳ always apply thermal compounds

Active Cooling

bigger fan = less noise

smaller fan = more noise

CPU Fan / CPU cooler

Case fan

Advanced CPU Cooling

Liquid Cooling

↳ regular maintenance